Lightweight Design of PMSM Motor Housing Using CFRP: A Comparative FEM Study

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Abstract

The growing demand for lightweight, high-efficiency electric vehicles (EVs) has led to increased focus on optimizing in dividual drivetrain components. Permanent Magnet Synchronous Motors (PMSMs), widely used in EVs due to their high efficiency and power density, rely on motor housings for structural support, protection, and alignment of internal components. However, conventional motor housings are typically manufactured from metallic materials such as steel and cast iron, which significantly contribute to the system's overall weight. This study investigates the feasibility of using Carbon Fiber Reinforced Polymer (CFRP) composites as an alternative mate rial for PMSM motor housing to achieve weight reduction without compromising structural performance. A composite laminate with a stacking sequence of [90/ + 45/0/ − 45]s was developed, and its effective mechanical properties were determined using Finite Element Method (FEM) analysis. These properties were then applied to a motor housing model, and a comparative structural analysis was conducted against conventional materials under identical loading conditions.

This study investigates the feasibility of using Carbon Fiber Reinforced Polymer (CFRP) composites as an alternative mate rial for PMSM motor housing to achieve weight reduction without compromising structural performance. A composite laminate with a stacking sequence of [90/ + 45/0/ − 45]s was developed, and its effective mechanical properties were determined using Finite Element Method (FEM) analysis. These properties were then applied to a motor housing model, and a comparative structural analysis was conducted against conventional materials under identical loading conditions.

The results demonstrate a significant weight reduction, with CFRP achieving approximately 4.5 times lower weight compared to aluminum. Additionally, the composite material exhibited lower stress levels, indicating improved load distribution. How ever, higher strain values and increased natural frequencies were observed, highlighting a trade-off in structural behavior. Overall, the

findings confirm that CFRP composites are a promising alter native for lightweight motor housing design in EV applications, offering substantial weight savings while maintaining structural integrity

1. Introduction

The rapid advancement of electric vehicles (EVs) has significantly increased demand for lightweight, high-performance components in modern drivetrain systems. Among motor technologies, Permanent Magnet Synchronous Motors (PMSMs) are widely adopted for their high efficiency, high power density, and compact size. These characteristics make PMSMs highly suitable for EV applications. While extensive research has been conducted on improving electromagnetic performance and thermal management, the structural aspects of the motor, particularly the motor housing, have received comparatively less attention.

The motor housing is a critical structural component that supports the stator assembly, maintains alignment of internal components, and protects the motor from external mechanical impacts. It also contributes to overall system stability and helps with heat dissipation. Traditionally, motor housings are manufactured using materials such as steel, cast iron, and aluminium due to their strength and manufacturability. However, these materials significantly increase the motor's overall weight, adversely affecting vehicle efficiency, energy consumption, and driving range.

To address these challenges, the automotive industry has placed greater emphasis on lightweight materials. Fiber reinforced polymer composites, particularly Carbon Fiber Reinforced Polymer (CFRP), offer high specific strength, low density, and excellent fatigue resistance. Additionally, composite materials provide the flexibility to tailor mechanical properties through Fiber orientation and laminate stacking sequences, enabling optimized structural performance under different loading conditions.

Despite these advantages, the application of CFRP composites in PMSM motor housings remains limited and warrants further investigation. There is a lack of comprehensive studies that combine composite laminate development with structural evaluation using the Finite Element Method (FEM). Therefore, this study focuses on developing a CFRP composite laminate, determining its effective mechanical properties, and applying these properties to a motor housing model for structural analysis.

A comparative FEM-based analysis is conducted between CFRP and conventional materials, including steel, cast iron, and aluminium. Key parameters, including deformation, stress, strain, and natural frequency, are evaluated to assess structural performance. The primary objective of this study is to investigate the feasibility of CFRP as a lightweight alternative material for PMSM motor housing, aiming to achieve significant weight reduction while maintaining structural integrity.

2. LITERATURE REVIEW

1. Lee and Lee investigated the structural design of a fully enclosed housing for interior PMSMs, emphasizing the importance of housing in protecting internal components and maintaining mechanical integrity. Their study highlights how housing design influences overall motor durability and operational stability.

2. Guo et al. focused on optimizing PMSM design for electric vehicle applications using a machine learning-based surrogate model integrated with FEM. Their work primarily addresses electromagnetic performance and design optimization, with limited consideration of structural components such as motor housings or material substitutions for weight reduction.

3. Devi presented a comprehensive review of electric motor modelling, analysis, and design for e-mobility applications. The study covers various aspects, including electromagnetic behavior and system-level design, providing a broad context for PMSM development. However, it identifies a gap in the detailed structural analysis of auxiliary components such as motor housings.

4. Kim et al. analyzed the vibration and noise characteristics of PMSMs, accounting for temperature-dependent material properties. Their work highlights the significance of structural and dynamic behavior in motor performance, particularly through FEM-based analysis.

5. Boopathi et al. investigated the application of lightweight composite materials for electric motor casings. They demonstrated that replacing conventional aluminium housings with composites can significantly reduce overall system weight while maintaining adequate structural performance. Their study highlights the potential of composites in improving energy efficiency and vehicle performance.

6. Dou et al. analyzed the vibration characteristics of CFRP laminates using FEM and experimental methods. The study revealed that CFRP structures exhibit higher natural frequencies and improved vibration damping compared to traditional materials, indicating their suitability for dynamic motor structural components.

7. Zu et al. examined the use of filament-wound CFRP composite sleeves in high-speed PMSM rotors. Their results showed that CFRP materials can effectively withstand high mechanical stresses while achieving significant weight reduction, although challenges related to stress distribution and manufacturing complexity were identified.

8. Shen et al. provided a comprehensive review of high speed PMSM design, emphasizing the importance of FEM based multiphysics analysis for evaluating structural strength, thermal behavior, and dynamic performance. Their work establishes the need for integrated analysis in advanced motor design.

9. Awan et al. reviewed thermal management strategies in PMSMs and highlighted that material selection plays a critical role in heat dissipation. The study noted that while lightweight materials are beneficial, composites such as CFRP require additional consideration due to their low thermal conductivity.

10. Mallick discussed the application of Fiber-reinforced composites in automotive structures, emphasizing their high strength-to-weight ratio and potential to reduce weight sig nificantly in structural components, making them suitable for replacing metals in vehicle parts.

11. Gay and Hoa explained the mechanical behavior of composite materials. They highlighted how laminate stacking sequences influence stiffness and strength, which are critical for designing load-bearing components such as motor housings.

12. Kaw presented fundamental concepts of composite materials, focusing on anisotropic behavior and the ability to tailor properties using Fiber orientation, which is essential for optimizing structural performance in engineering applications.

13. Gibson analyzed the mechanical behavior of composite materials and emphasized their application in lightweight structural design, including their superior fatigue resistance and adaptability for complex geometries.

14. Daniel and Ishai studied the engineering mechanics of composite materials. They demonstrated how FEM can be used to evaluate stress and strain behavior in composite laminates, supporting their use in structural applications.

15. Li et al. investigated the structural design of electric motor housings using FEM, focusing on stress distribution, deformation, and weight reduction. Their study showed that optimizing housing design and material selection improves structural performance and reduces weight.

3. GEOMETRIC DESCRIPTION

A. Geometric Description of Designed Laminate

The geometry considered in this study is a square composite laminate model generated in ANSYS SpaceClaim. The plate has overall dimensions of 1m x 1m, i.e., 1000mm x 1000mm, with a uniform total thickness of 1mm. The model is created as a 2D planar surface, which is appropriate for thin composite structures where the thickness is significantly smaller compared to the in-plane dimensions. The thickness is incorporated through layered composite definitions rather than explicit 3d solid modelling.

The laminate is defined using a symmetric stacking sequence given by: $[90/+45/0/-45]_s$.

This configuration consists of eight plies (stacked), each with a thickness of 0.125mm, resulting in a total laminate thickness of 1 mm. The symmetric arrangement ensures balanced mechanical behavior and eliminates bending extension coupling. Each layer is assigned Epoxy Carbon UD (230 GPa) material with the specified Fiber orientation relative to the global coordinate system. The Fiber direction in each ply is defined with respect to the global X-axis, which represents the principal material direction.

A global Cartesian Coordinate system is used to define the geometry:

- X-axis: along the plate length
- Y-axis: along the plate width
- Z-axis: through the thickness direction

The geometry is discretized using a uniform mesh with an element size of 25 mm, ensuring adequate resolution to capture the laminate's structural response.

B. Geometric Description of PMSM Motor Housing

The motor housing is a cylindrical structure that encloses and supports the internal components of a Permanent Magnet Synchronous Motor (PMSM). The geometry consists of a hollow cylindrical body with integrated end features for mounting and alignment.

The main housing has an outer diameter of approximately 117.74 mm and an overall length of about 217 mm. The inner cavity is designed to accommodate the stator assembly, with an

internal length of around 155 mm and a uniform wall thickness of approximately 10 mm. The cylindrical body features external longitudinal fins along its surface, enhancing structural stiffness and potentially aiding heat dissipation.

At one end, the housing features a flange with a thickness of about 7 mm and a rounded edge profile (R10), providing structural reinforcement and mounting support. The opposite end features a stepped geometry with a reduced-diameter sec tion (approximately 30–35 mm), designed for shaft alignment and bearing placement.

Detailed sectional views show internal shoulders and grooves that aid component location and ensure proper assembly. Fillets (R5–R10) are incorporated at critical junctions to reduce stress concentration. Additional features, such as small cutouts and slots, are present in the end sections, likely for ventilation, weight reduction, or assembly.

The housing also includes multiple concentric diameter steps (e.g., 90 mm, 100 mm, 107.63 mm), indicating transitions for fitting internal components. The end cap region has a thickness of approximately 3 mm and angular features (∼38◦), which contribute to both structural integrity and manufacturability.

Overall, the geometry is designed to balance strength, weight, and functionality, ensuring proper support, alignment, and protection of motor components while enabling efficient operation.

4. SIMULATION AND ANALYSIS

A. Laminate

A square laminate, 1000mm in length and 1000mm in width, with a total thickness of 1mm, was tested in static structural analysis under controlled loading conditions in An sys Workbench. A uniform mesh size of 25mm was used to evaluate orthotropic properties.

To evaluate the longitudinal elastic modulus Ex, a tensile load of 1000N was applied along the x-axis at one edge, and the opposite edge was fully constrained to prevent body motion. Normal stress and corresponding strain were obtained. We used the linear elastic relation to obtain E_x.
Average normal stress = 0.15021 MPa
Average normal strain = 0.000020654

$$E_x = \frac{\text{Stress}}{\text{Strain}} = 7276.9\,\text{MPa} = 7.28\,\text{GPa}$$

The transverse elastic modulus Ey was determined by stress-strain response in the y-direction under similar loading conditions.
Average stress = 0.0020231 MPa
Average strain = 0.0000053178

$$E_y = \frac{\text{Stress}}{\text{Strain}} = 123360\,\text{MPa} = 123.4\,\text{GPa}$$

Then, to evaluate the Shear Modulus, we applied shear stress along the opposite edges, parallel to the y-axis.

Shear stress = 8.2557×10^{-21}

Shear strain = 1.0903×10^{-15}

$$G_{xy} = \frac{\tau_{xy}}{\gamma_{xy}} = 5000\,\text{MPa} = 5\,\text{GPa}$$

By applying a tensile load on opposite edges, a force of 1000N was applied on the x-axis on one edge, and the opposite edge was fully constrained to prevent motion of the body. Normal elastic strain along the x and y directions was obtained and used to calculate Poisson's ratio.

Axial strain = 0.000020654

Transverse strain = −0.0000053178

$$\nu_{xy} = -\frac{\text{Transverse strain}}{\text{Axial strain}} = 0.25541 \approx 0.255$$

$$\nu_{yx} = \nu_{xy} \times \left(\frac{E_y}{E_x}\right) = 0.015$$

TABLE 1

EPOXY CARBON 230 GPA WET COMPOSITE LAMINATE PROPERTIES

Property	Value
Longitudinal Modulus, E_x	7.28 GPa
Transverse Modulus, E_y	123.4 GPa
Shear Modulus, G_{xy}	5 GPa
Poisson's Ratio, ν_{xy}	0.255
Reciprocal Poisson's Ratio, ν_{xy}	0.015

B. Motor Housing

The PMSM motor housing is tested using the Finite Element Method (FEM), accounting for combined mechanical and thermal loading conditions. The 3D CAD model of the motor housing is first imported into the analysis software and simplified where necessary while retaining all critical structural features such as the bore, mounting zones, and flange regions. Appropriate material properties are then assigned based on the selected material, including anisotropic properties in the case of CFRP composites.

The model is discretized using appropriate finite elements, and a mesh convergence study is conducted to verify the accuracy of the results. A finer mesh is applied in regions prone to stress concentration, such as fillets and mounting areas. Boundary conditions are applied by fixing all four mounting zones with zero degrees of freedom, representing actual assembly constraints.

Loading conditions are then applied to simulate real operating scenarios. A torque of 20 Nm is applied on the inner bore surface to represent shaft-induced loading. The self-weight of the housing is calculated using the acceleration due to gravity. In addition, thermal loading is introduced by assigning an operating temperature of 80◦C to the motor region and an ambient temperature of 26◦C, creating a temperature gradient that induces thermal stresses in the structure.

A static structural analysis is performed to evaluate de formation, stress, and strain under these combined loading conditions. The results are post-processed to identify the max imum deformation, equivalent (von Mises) stress, and strain distribution, with particular focus on critical regions, including the bore, mounting zones, and fillet areas. The obtained results are then compared with allowable material limits to assess structural safety and overall performance of the motor housing.

5. MATERIALS, PROPERTIES, AND COMPARISON

In this study, the structural performance of the PMSM motor housing is evaluated using both conventional metallic and advanced composite materials. The selected materials include steel, cast iron, aluminum, and Carbon Fiber Reinforced Polymer (CFRP). These materials are commonly used or considered in automotive structural applications due to their mechanical strength and manufacturability.

Steel and cast iron are widely used in motor housings due to their high strength and stiffness. Steel offers excellent mechanical properties, including high yield strength and durability, making it suitable for heavy-duty applications. Cast iron offers good compressive strength and vibration-damping properties, but is relatively heavy. Aluminum is preferred in electric vehicle applications because of its lower density compared to steel and cast iron, along with adequate strength and good corrosion resistance. However, even aluminum contributes significantly to the motor housing’s overall weight.

To overcome the limitations of metallic materials, CFRP composites are considered an alternative. CFRP offers a high strength-to-weight ratio, low density, and excellent fatigue

resistance. Unlike isotropic metals, CFRP exhibits anisotropic behavior, meaning its mechanical properties vary with fiber orientation. In this study, a laminate stacking sequence of [90 / +45 / 0 /-45].

Table 1 : Stacking

Layer	Material	Thickness(mm)	Angle(°)
(+Z)			
8	Epoxy Carbon UD (230 GPa) Wet	0.125	90
7	Epoxy Carbon UD (230 GPa) Wet	0.125	45
6	Epoxy Carbon UD (230 GPa) Wet	0.125	0
5	Epoxy Carbon UD (230 GPa) Wet	0.125	-45
4	Epoxy Carbon UD (230 GPa) Wet	0.125	-45
3	Epoxy Carbon UD (230 GPa) Wet	0.125	0
2	Epoxy Carbon UD (230 GPa) Wet	0.125	45
1	Epoxy Carbon UD (230 GPa) Wet	0.125	90
(-Z)			

Table 2 : Material Comparison

Property	Structural Steel	Cast Iron	CFRP	Aluminium Alloys
Density (kg/m^3)	7850	7100-7300	1500-1600	2700
Young's Modulus, E (GPa)	200	110-170	70-140	69-72
Shear Modulus, G (GPa)	80	45-70	4-6 (in-line)	25-27
Poission's Ratio (v)	0.3	0.25-0.28	0.25-0.35	0.33
Ultimate Tensile Strength (MPa)	400-500	150-300	600-1500 (Fiber dir.)	200-600
Relative Stiffness	High	Medium	Direction-dependent	Medium
Weight (kg)	6.776	6.336	1.408	2.376
Relative Cost	Low	Low-Medium	High	Medium

The evaluated composite properties include longitudinal modulus (Ex = 7.28 GPa), transverse modulus (Ey = 123.4 GPa), shear modulus (Gxy = 5 GPa), and Poisson's ratios (vxy = 0.255 and vyx ≈ 0.015). These values indicate strong directional stiffness, allowing the material to be tailored to loading requirements. Compared with conventional materials, CFRP has a significantly lower density, resulting in substantial weight reduction. The study observed a reduction in weight of approximately 4.5 times.

From a structural perspective, CFRP shows lower stress values under similar loading conditions, indicating better stress distribution. However, higher strain values are observed due to its relatively lower stiffness in certain directions. Additionally, modal analysis reveals higher natural frequencies in CFRP, which can influence the motor housing's vibration characteristics.

Overall, the comparison highlights that while steel and cast iron provide high stiffness and strength, they are limited by their high weight. Aluminum offers a balance between weight and strength, but is still heavier than composites. CFRP emerges as a promising alternative due to its lightweight nature and adequate structural performance, although considerations such as anisotropy and thermal conductivity must be addressed for practical applications.

6. METHODOLOGY

The methodology adopted in this study involves a system atic approach to develop, analyze, and compare the structural performance of different materials for PMSM motor housing using the Finite Element Method (FEM). The entire process is divided into sequential stages, starting from composite development to final result evaluation.

Initially, a composite laminate is developed using Carbon Fiber Reinforced Polymer (CFRP). A symmetric stacking sequence of [90 / +45 / 0 /-45]s is selected to achieve balanced mechanical properties in multiple directions. This layup configuration ensures improved performance under complex loading conditions by distributing stiffness and strength across different orientations.

Following laminate development, the composite's effective mechanical properties are determined. A representative lami nate model is created and analyzed using FEM to extract en gineering constants, including the longitudinal modulus (Ex), transverse modulus (Ey), shear modulus (Gxy), and Poisson's ratios (νxy and νyx). These properties capture the composite's anisotropic nature and are later used as input material data for structural analysis.

Subsequently, the geometric model of the PMSM motor housing is prepared using appropriate CAD tools. The model is simplified where necessary to ensure computational efficiency while retaining critical structural features. After geometry preparation, the model is imported into the FEM environment for analysis. In the next stage, the finite element setup is carried out. This includes defining material properties for each material (steel, cast iron, aluminum, and CFRP), applying boundary conditions, and assigning loading conditions repre sentative of operational scenarios. A mesh convergence study is performed to determine an optimal mesh size that yields accurate results at a reasonable computational cost.

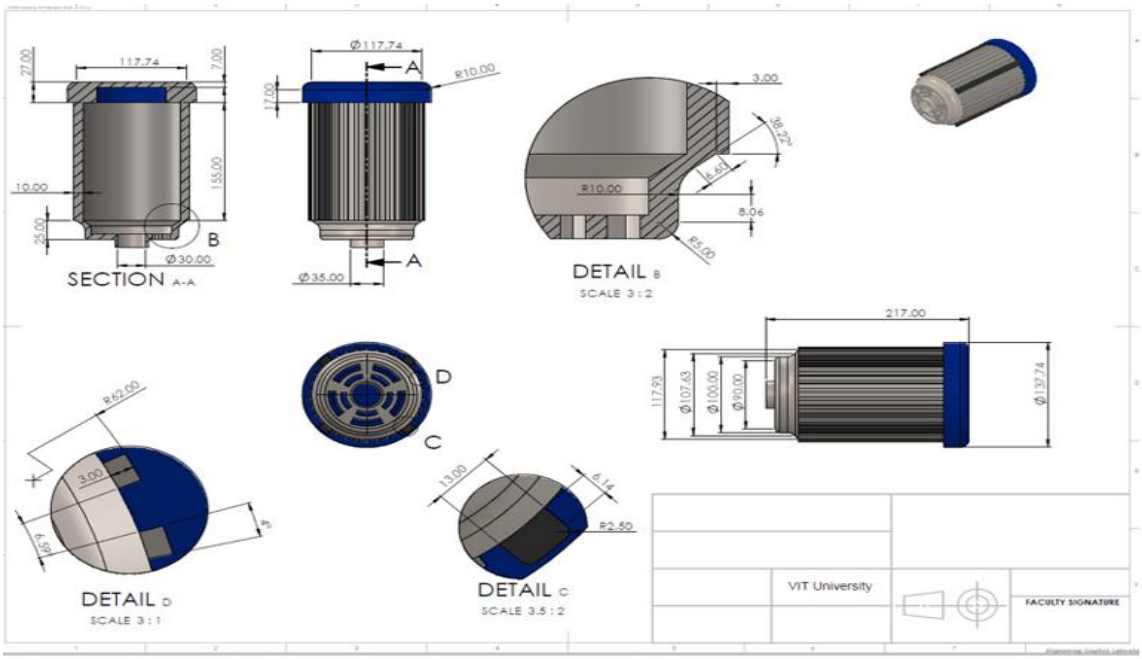


Fig.1 :- Drawing Sheet (Motor Housing)

Once the model setup is complete, structural analysis is performed. Key parameters, including total deformation, equivalent (von Mises) stress, and strain distribution, are evaluated for each material. In addition to static structural analysis, modal analysis is conducted to determine the natural frequencies of the motor housing, which are important for understanding vibration characteristics.

Finally, the results obtained from FEM analysis are eval uated and compared across all materials. The performance of CFRP is assessed against conventional materials in terms of weight, stress distribution, deformation, strain, and natural frequency. This comparative evaluation is used to determine the feasibility of CFRP as a lightweight alternative for PMSM motor housing.

A. Governing Equations

The structural and modal behavior of the motor housing is governed by the finite element formulation. The global equilibrium equation relates the applied forces to the nodal displacements through the stiffness matrix:

$$[K]\{u\}=\{F\} \qquad (1)$$

where [K] is the global stiffness matrix, {u} is the displace ment vector, and {F} is the external force vector.

The strain developed in the structure is related to displace ment through the strain-displacement matrix:

$$\{\varepsilon\} = [B]\{u\} \qquad (2)$$

where {ε} is the strain vector and [B] is the strain displacement matrix.
The stress-strain relationship follows Hooke's law for linear elastic materials:

$$\{\sigma\} = [D]\{\varepsilon\} \quad (3)$$

where {σ} is the stress vector and [D] is the material stiffness matrix. For composite materials such as CFRP, the matrix [D] represents anisotropic behavior.

Combining the above relations, the stress can be expressed in terms of displacement:

$$\{\sigma\} = [D][B]\{u\} \quad (4)$$

For dynamic analysis, the natural frequencies of the structure are obtained from the eigenvalue problem:

$$([K] - \omega^2[M])\{\phi\} = 0 \quad (5)$$

where [M] is the mass matrix, ω is the natural frequency and {ϕ} is the mode shape vector. For a simplified system, the natural frequency can also be expressed as:

$$\omega = \sqrt{\frac{k}{m}} \quad (6)$$

where k is the stiffness and m is the mass of the system.

7. RESULTS AND DISCUSSION

The structural and modal analysis of the PMSM motor housing was carried out using Finite Element Method (FEM) for different materials, including steel, cast iron, aluminum, and Carbon Fiber Reinforced Polymer (CFRP). The results obtained provide a comprehensive comparison of deformation, stress distribution, strain behavior, and natural frequency, enabling evaluation of the feasibility of CFRP as a lightweight alternative.

The deformation analysis indicates that CFRP exhibits slightly higher deformation compared to conventional metallic materials. This behavior is attributed to the composite's anisotropic nature and its lower stiffness in certain directions. However, the observed deformation remains within acceptable limits, confirming that the motor housing's structural integrity is not compromised. The stress analysis shows that CFRP experiences lower equivalent (von Mises) stress values than steel, cast iron, and aluminum under identical loading conditions. This suggests that the composite material provides better stress distribution and reduced stress concentration, both of which are beneficial for structural durability and fatigue performance.

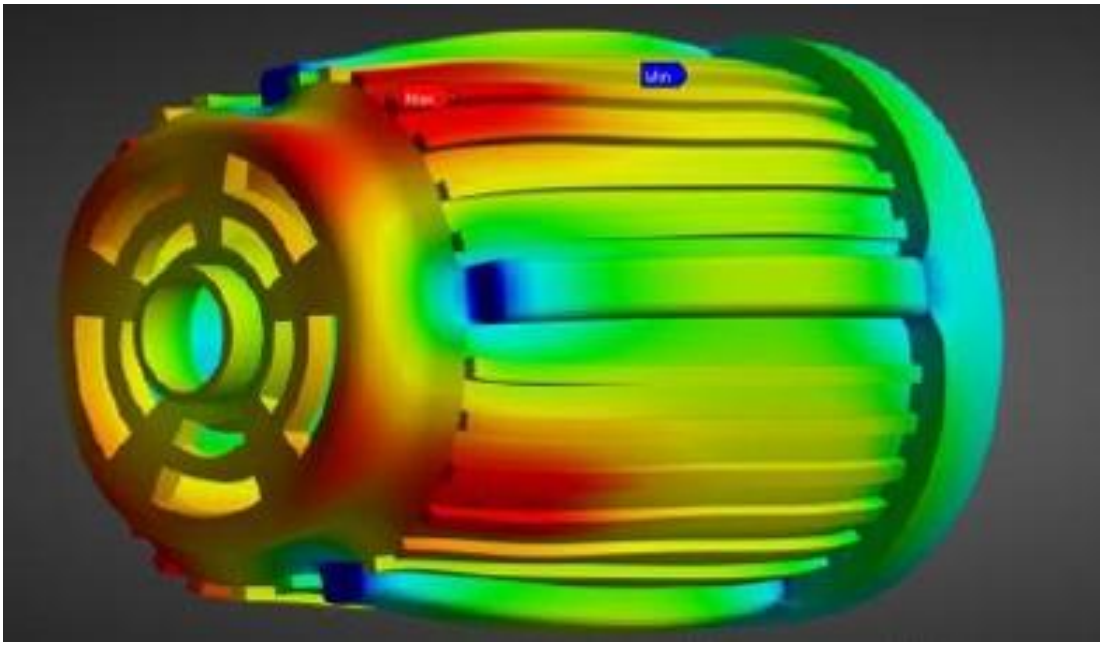


Fig. 2: Displacement Schematic

Table 3: Displacement Comparison

Material	Min. Displacement (m)	Max. Displacement (m)	Deviation (%)
Alloy Steel	0	7.96E-05	5.40E+01
Aluminium	0	1.73E-04	1.00E+02
Cast Iron	0	7.26E-05	4.20E+01
CFRP	0	1.86E-04	1.08E+02

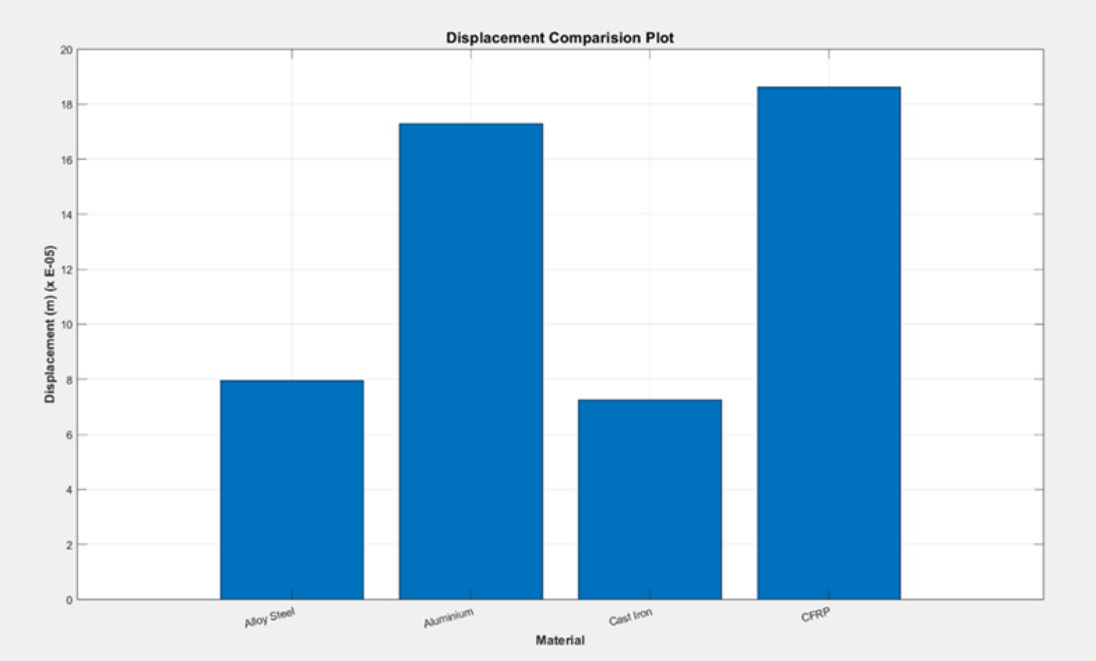


Fig. 2: Displacement Comparison

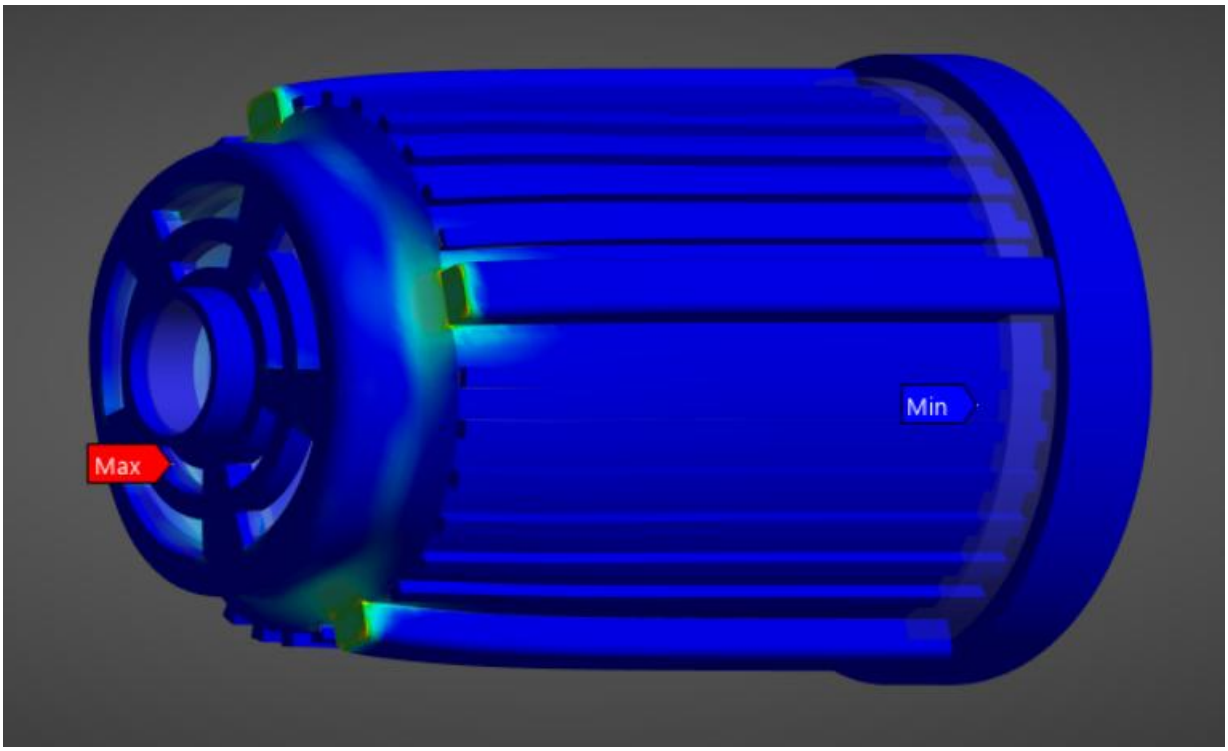


Fig. 3: Stress Schematic

Table 4 : Stress Comparison

Material	Min. Stress (Pa)	Max. Stress (Pa)	Deviation (%)
Alloy Steel	8.73E+05	1.69E+09	4.87E+01
Aluminium	2.06E+05	1.14E+09	1.00E+02
Cast Iron	4.57E+05	8.61E+08	7.56E+01
CFRP	51419	2.46E+08	2.16E+01

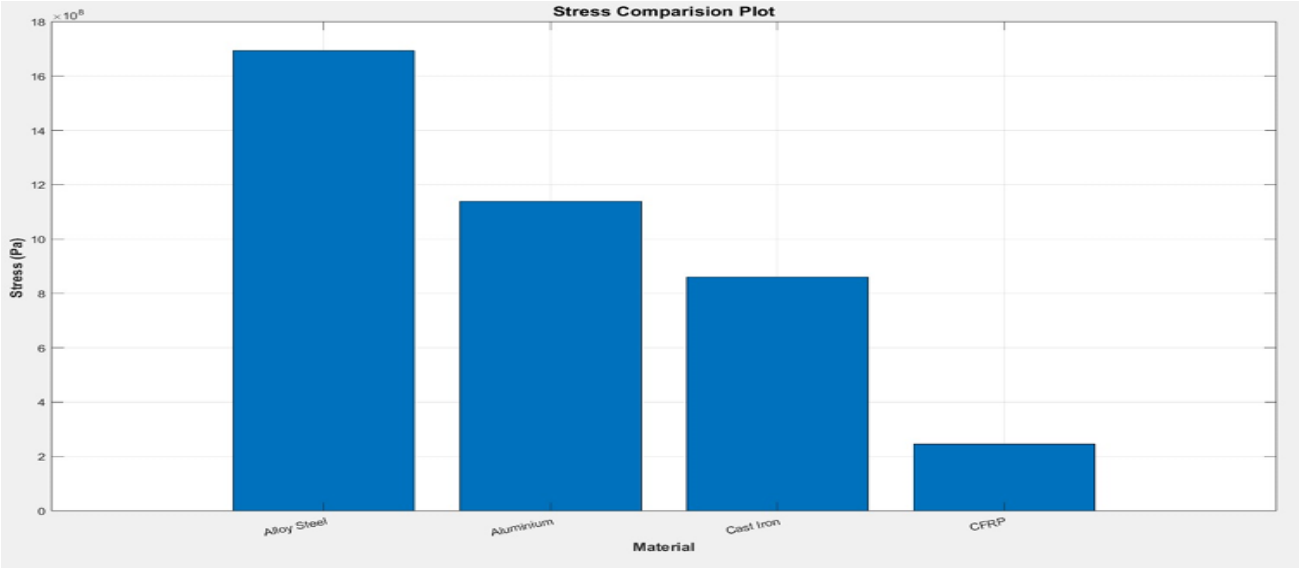


Fig. 4 : Stress Comparison Graph

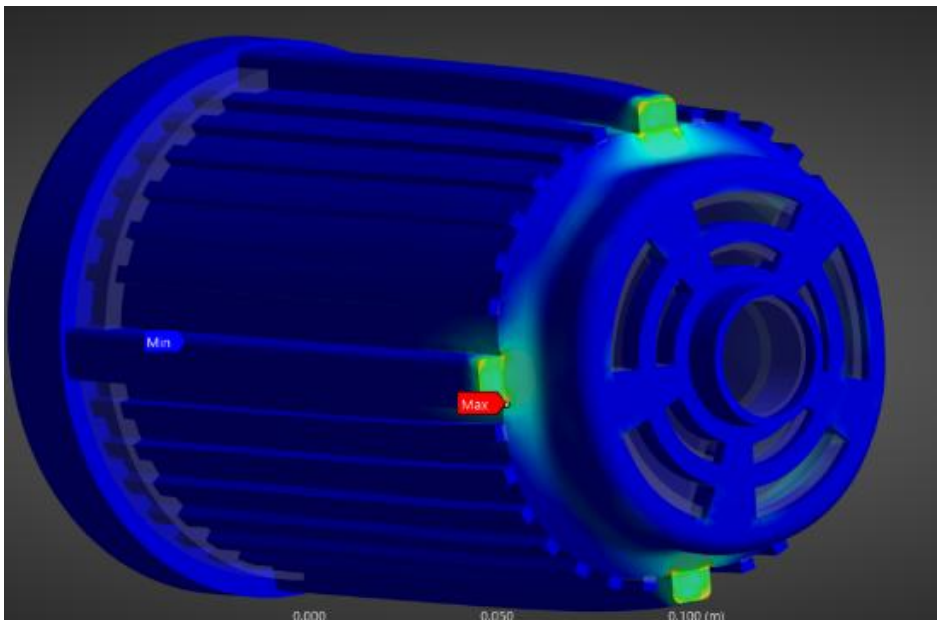


Fig. 5: Strain Schematic

Table 5 : Strain Comparison

Material	Min. Strain	Max. Strain	Deviation(%)
Alloy Steel	5.95E-06	1.30E-02	4.85E+01
Aluminium	6.92E-06	2.52E-02	1.00E+02
Cast Iron	5.87E-06	1.18E-02	4.68E+01
CFRP	8.28E-06	3.94E-02	1.56E+02

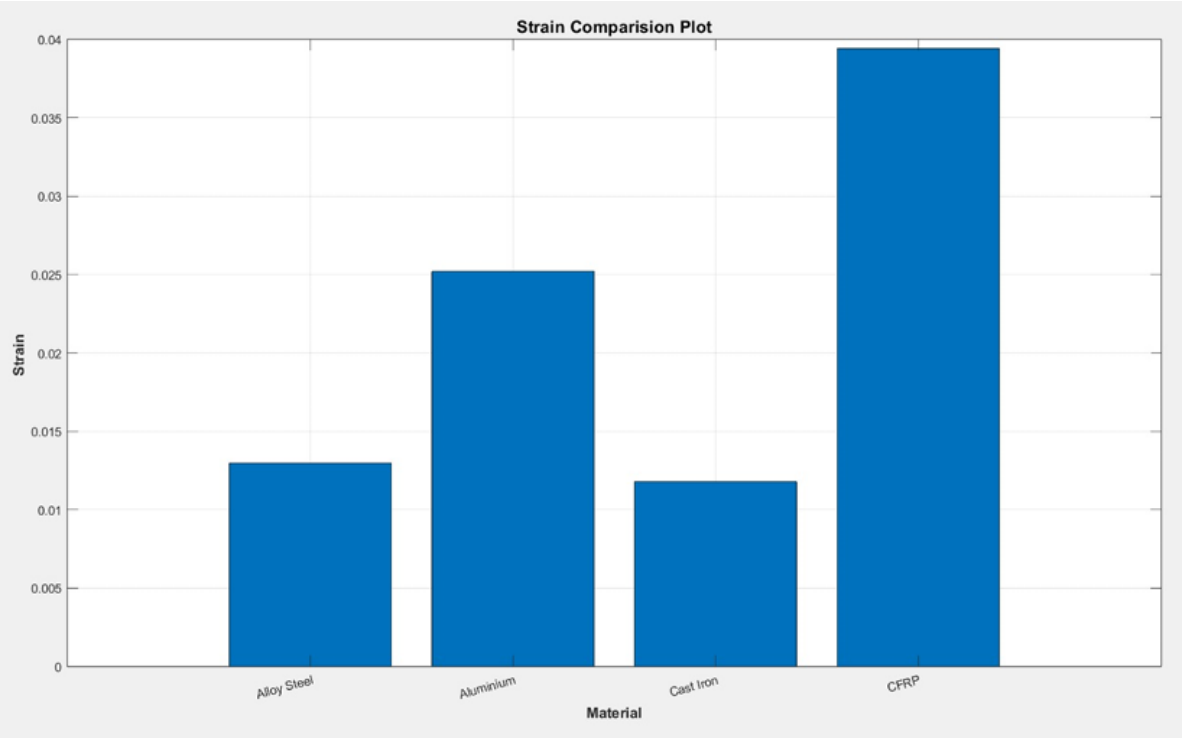


Fig. 6 : Stress Comparison Graph

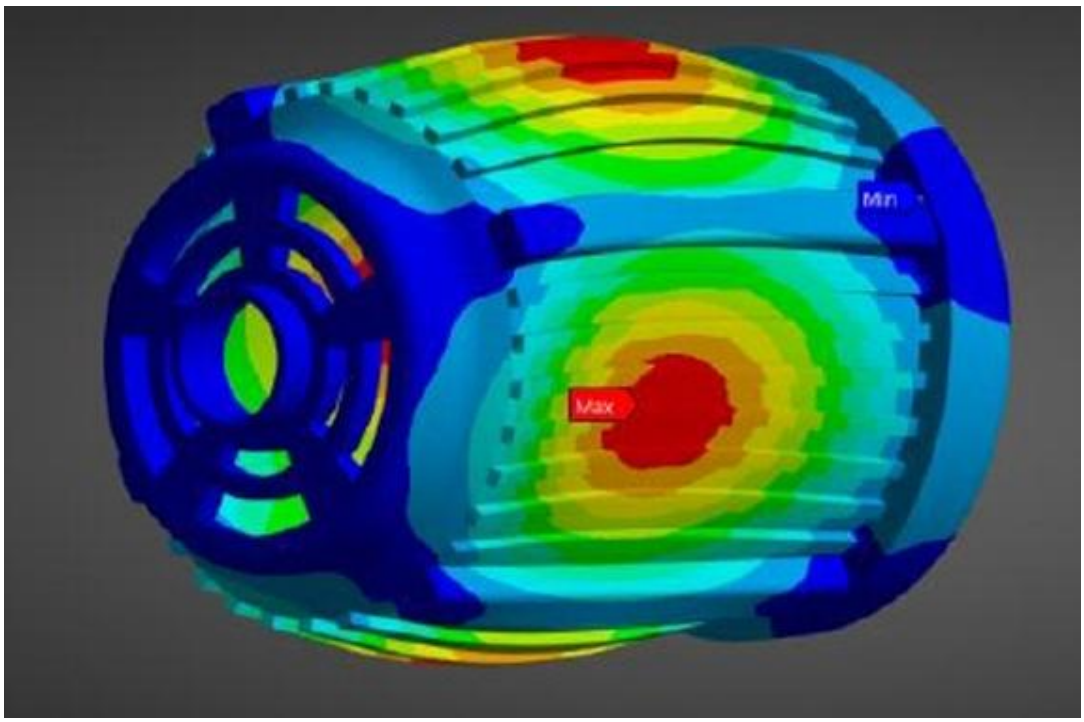


Fig 7 : Modal Analysis

Table 6 : Natural Frequency Comparison

Material	Natural Frequency (Hz)	Deviation (%)
Alloy Steel	3.60E+03	3.81E-01
Aluminium	3.62E+03	1.00E+02
Cast Iron	2.79E+03	7.71E+01
CFRP	1.97E+03	5.45E+01

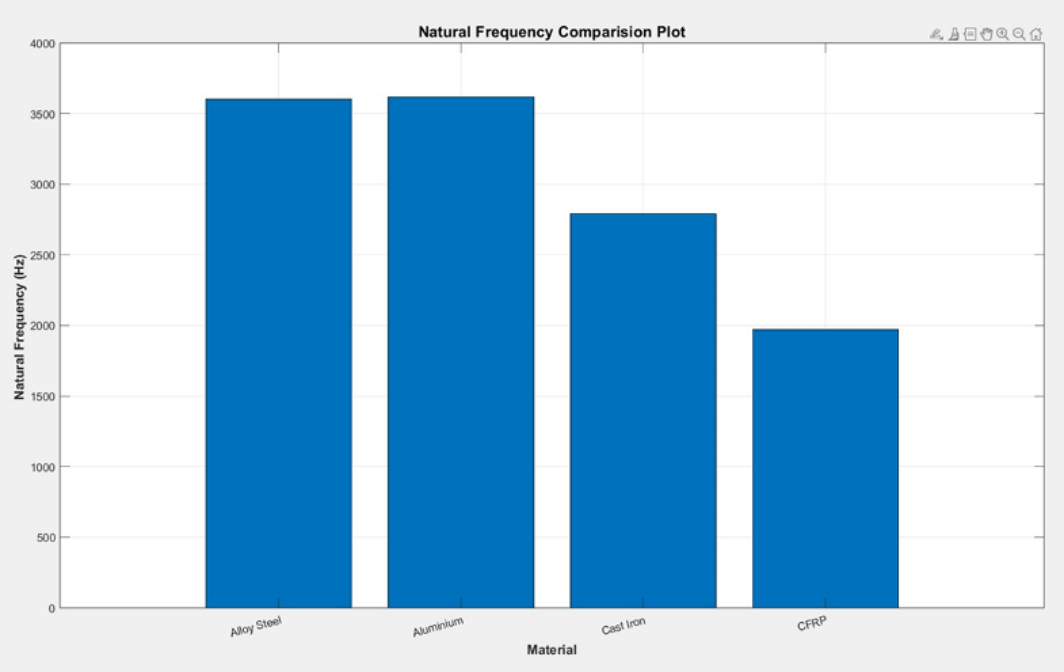


Fig. 8 : Natural Frequency Comparison Graph

In terms of strain behavior, CFRP demonstrates higher strain values than conventional materials. This is expected due to its directional mechanical properties and lower modulus in specific orientations. Although higher strain may indicate increased deformation under load, it also reflects the material's ability to absorb energy without failure, which can be advantageous in dynamic conditions. A significant outcome of this study is the substantial weight reduction achieved with CFRP. The results show a weight reduction of approximately 4.5 times compared to aluminium, highlighting the effectiveness of composite materials in lightweight design. This reduction directly

contributes to improved efficiency, reduced energy consumption, and enhanced driving range in electric vehicles.

Modal analysis shows that CFRP exhibits higher natural frequencies than metallic materials. A higher natural frequency indicates an improved stiffness-to-mass ratio and reduced susceptibility to resonance under operating conditions. This characteristic helps minimize vibration-related issues during motor operation.

Overall, the results demonstrate that CFRP provides a favourable balance between weight reduction and structural performance. While there are trade-offs, such as increased deformation and strain, the benefits of reduced stress and significant weight savings make CFRP a promising alternative for PMSM motor housing applications. The findings validate the potential of composite materials for advanced lightweight design in electric vehicle systems.

8. CONCLUSION

The present study investigates the feasibility of using Carbon Fiber Reinforced Polymer (CFRP) as an alternative material for the PMSM motor housing through a comparative Finite Element Method (FEM) analysis with conventional materials, including steel, cast iron, and aluminium. The results demonstrate that CFRP offers a significant advantage in weight reduction, achieving approximately 4.5 times lower weight than aluminium. This substantial reduction contributes directly to improved efficiency, lower energy consumption, and enhanced performance of electric vehicles.

From a structural perspective, CFRP exhibits lower stress values under identical loading conditions, indicating better stress distribution and reduced likelihood of failure. Although higher deformation and strain values are observed, they remain within acceptable limits and reflect the inherent anisotropic and energy-absorbing nature of composite materials. Addition ally, the higher natural frequency observed in CFRP suggests improved vibration characteristics and reduced resonance issues during operation.

Overall, the study confirms that CFRP is a promising lightweight alternative for PMSM motor housing, capable of maintaining structural integrity while significantly reducing weight. The use of tailored laminate stacking sequences further enhances its adaptability for specific loading conditions, making it suitable for advanced automotive applications.

However, certain drawbacks and limitations must be considered. CFRP has relatively low thermal conductivity compared to metals, which may affect heat dissipation in motor applications and require additional thermal management solutions. The anisotropic nature of composites also makes design and analysis more complex compared to isotropic materials. Manufacturing processes for CFRP are generally more expensive and require specialized techniques, thereby increasing production costs. Additionally, issues related to recyclability and end-of-life disposal of composite materials pose environmental challenges.

In conclusion, while CFRP provides significant advantages in lightweight design and structural performance, addressing challenges related to thermal behavior, cost, and sustainability is essential for its large-scale implementation in PMSM motor housing applications.

9. FUTURE SCOPE

The present study establishes the structural feasibility of Carbon Fiber Reinforced Polymer (CFRP) for PMSM motor housing; however, several areas remain open for further research and development to enable practical implementation in electric vehicle applications.

One important area for future work is thermal management. Since CFRP has relatively low thermal conductivity compared to conventional metals, a detailed thermal analysis is required to evaluate heat dissipation from the motor. Advanced cooling techniques or hybrid material designs may be explored to overcome this limitation.

Another significant direction is the Life Cycle Assessment (LCA) of CFRP motor housing. Evaluating the environmental impact, recyclability, and sustainability of composite materials throughout their lifecycle will be essential for large-scale adoption in the automotive industry.

Further studies can also focus on optimizing the composite laminate design. Different stacking sequences, fiber orientations, and layer thicknesses can be investigated to achieve an optimal balance between stiffness, strength, and weight. This can lead to improved structural performance tailored to specific loading conditions.

Experimental validation of the FEM results is another crucial step. Physical testing of prototype CFRP motor housings can help verify simulation results and ensure real-world reliability under operating conditions.

Additionally, future research may explore hybrid materi als that combine composites with metals to leverage both lightweight properties and improved thermal conductivity. Manufacturing techniques and cost optimization strategies can also be investigated to make CFRP more economically viable for mass production.

Overall, these future directions will help overcome current limitations and enhance the practical applicability of CFRP in the design of PMSM motor housings for next-generation electric vehicles.

APPENDIX A: GOVERNING EQUATIONS FOR STRUCTURAL AND MODAL ANALYSIS

This appendix presents the fundamental equations used in the finite element analysis of the PMSM motor housing. These equations govern the structural response and dynamic behavior of the system.

A. Displacement Equation

The global equilibrium equation relates the applied forces to nodal displacements:

$$[K]\{u\} = \{F\} \quad (7)$$

where [K] is the global stiffness matrix, {u} is the displacement vector, and {F} is the external force vector.

B. Strain–Displacement Relation

The strain developed in the structure is related to displacement as:

$$\{\varepsilon\} = [B]\{u\} \quad (8)$$

where {ε} is the strain vector and [B] is the strain displacement matrix.

C. Stress–Strain Relation

The stress-strain relationship for linear elastic materials is given by:

$$\{\sigma\} = [D]\{\varepsilon\} \quad (9)$$

D. Stress–Displacement Relation

By combining the above equations, stress can be expressed in terms of displacement:

$$\{\sigma\} = [D][B]\{u\} \quad (10)$$

E. Natural Frequency Equation

The natural frequencies of the structure are obtained from the eigenvalue problem:

$$([K] - \omega^2[M])\{\phi\} = 0 \quad (11)$$

where [M] is the mass matrix, ω is the natural frequency and {ϕ} is the mode shape vector.

F. Simplified Natural Frequency

For a simplified system, the natural frequency can be expressed as:

$$\omega = \sqrt{\frac{k}{m}} \quad (12)$$

where k is the stiffness and m is the mass of the system.

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